# Predicting Falcon 9 First Stage Landing Success

A Data-Driven Analysis for Launch Cost Estimation




© IBM Corporation. All rights reserved.



By Rhyz Manalo

16th October 2026

# OUTLINE



- Executive Summary
- Introduction
- Methodology
- Results
  - Visualisation – Charts
  - Folium Maps & Markers
  - Dashboard
  - Predictive Analysis
- Discussion
- Conclusion
- Appendix

# EXECUTIVE SUMMARY



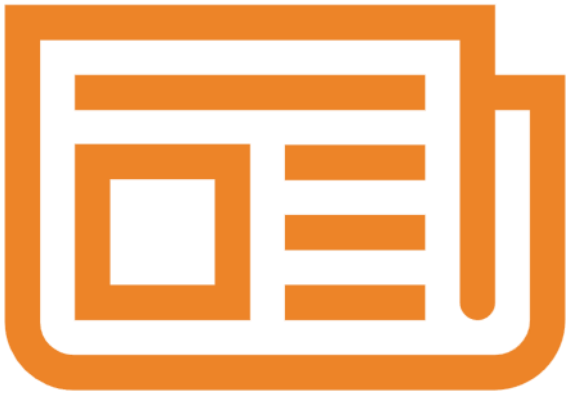
- Project Objective: Use Machine Learning model that predicts the successful landing of a Falcon 9 (first stage), enabling more accurate launch cost estimation.
- Methodology:
  1. Data Wrangling: Collected & cleaned historical launch data using SpaceX APIs & Web Scraping.
  2. Exploratory Data Analysis: Analysed factors influencing landing success, including launch site, payload mass, and orbit type.
  3. Interactive Visual Analytics: Created dashboards + maps to identify patterns & optimise launch site selection.
  4. Predictive Modelling: Trained & tuned four classification models to find the most accurate predictor (Logistic Regression, SVM, Decision Trees, KNN).
- Key Findings:
  - Falcon 9's landing success significantly increased over time.
  - Launch Sites: Success rates vary depending on location, with KSC LC-39A showing the best performance.
  - Best Model: The Decision Tree Model had the highest accuracy (83.3%)
- Recommendations:
  - Use the Decision Tree model to pre-launch predictions
  - This model can provide reliable cost estimates for potential clients, improving financial planning.

# INTRODUCTION



- Problem: SpaceX launch cost depends on whether the Falcon 9 first stage lands successfully and can be reused.
- Goal: Predict first-stage landing outcomes
- Key Questions:
  - Which factors influence landing success?
  - Which launch sites and orbits perform best?
  - Can machine learning accurately predict landing?
- Scope: Falcon 9 only (historical launch data)

# METHODOLOGY: Part 1



- Data Sources:
  - SpaceX API: https://api.spacexdata.com/v4/launches/past/
  - Wikipedia Falcon 9 Launch History Tables
- Data Wrangling:
  - Parsed JSON/HTML => Pandas Dataframe
  - Sampled out Falcon 1 Data
  - Filled Payload Mass Null Values with the Mean
  - Launch Pad Null Values Kept (Represents No Lauch Pad Usage)
- Feature Engineering:
  - Mapped 8 Outcomes => Binary Y (0: Bad, 1: Good)
  - Used launch pad numbers, site, payload mass, booster version, etc.
- Analysis & Modelling:
  - EDA, Folium Maps, Plotly Dash, Dash Dashboard
  - Models: Logistic Regression, SVM, Decision Trees, KNN
  - GridSearch + Train Test Split + Confusion Matrix

# EDA: Using SQL To Extract Relevant Extract Data

- Magic Commands: After Data Collection from relevant APIs and Web Scraping, we must prepare the data for analysis. Here are some example queries used:
- Remove Blank Spaces from Table:
  - %sql DROP TABLE IF EXISTS SPACEXTABLE;
  - %sql create table SPACEXTABLE as select * from SPACEXTBL where DATE is not null
- List Total Number of Successful & Failure Mission Outcomes:
  - %sql SELECT Misson_Outcome, COUNT (*) AS Total FROM SPACEXTABLE GROUP BY Mission_Outcome;
- Rank the Count of Landing Outcomes (Failure (Drone Ship) or Success (Ground Pad) between 2010-06-04 and 2017-03-20, in Descending order):
  - %sql SELECT Landing_Outcome, Count (*) AS COUNT FROM SPACEXTABLE WHERE Date BETWEEN '2010-06-04' ABD '2017-03-20' GROUP BY Landing_Outcome ORDER BY Count DESC;

# EDA: SQL Queries Results

| Booster_Version |
|---|
| F9 FT B1022 |
| F9 FT B1026 |
| F9 FT B1021.2 |
| F9 FT B1031.2 |

List Names of the Boosters Which have Success in Drone Ship & have Payload Mass Greater than 4000 but less than 6000

List Total Number of Successful & Failure Mission Outcomes:

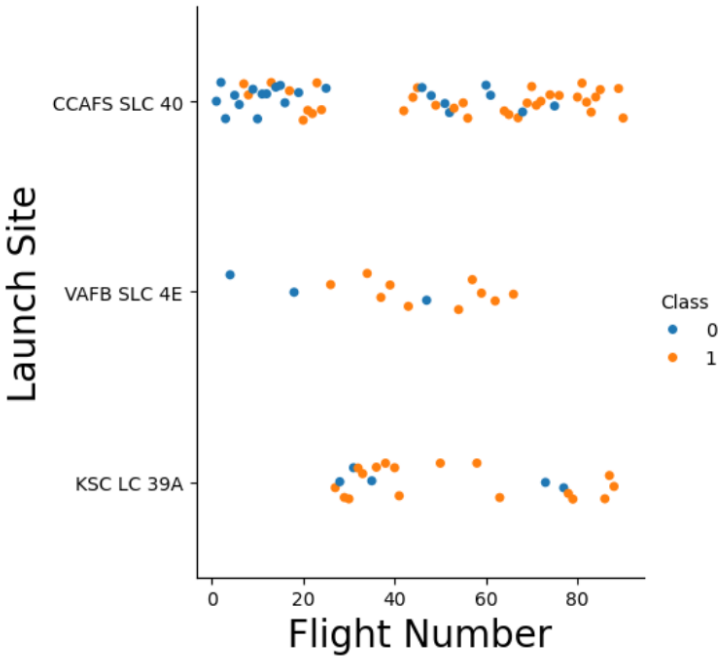
| Mission_Outcome | Total |
|---|---|
| Failure (in flight) | 1 |
| Success | 98 |
| Success | 1 |
| Success (payload status unclear) | 1 |

| Landing_Outcome | Count |
|---|---|
| No attempt | 10 |
| Success (drone ship) | 5 |
| Failure (drone ship) | 5 |
| Success (ground pad) | 3 |
| Controlled (ocean) | 3 |
| Uncontrolled (ocean) | 2 |
| Failure (parachute) | 2 |
| Precluded (drone ship) | 1 |

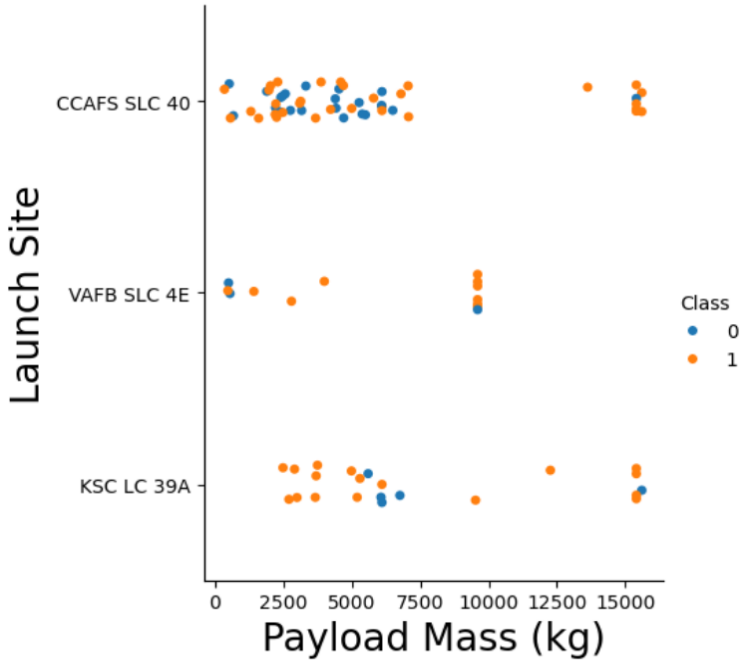
Rank the Count of Landing Outcomes (Failure (Drone Ship) or Success (Ground Pad) between 2010-06-04 and 2017-03-20, in Descending order):

# RESULTS: Launch Site Trends in Flight Numbers & Payload Mass (kg)

Takeaway: Success Rate Significantly Increases with Flight Number




Scatter Plot showing Flight Number vs Launch Site (Class)



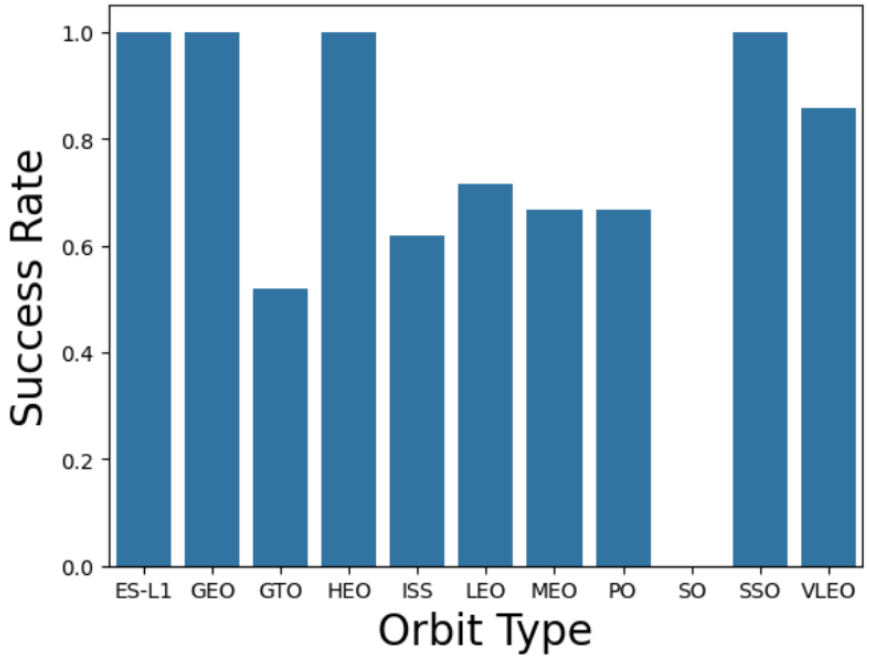

Scatter Plot showing Payload Mass (kg) vs Launch Site (Class)

Takeaway: Success Rate Varies with Payload Mass (kg) for Different Launch Sites

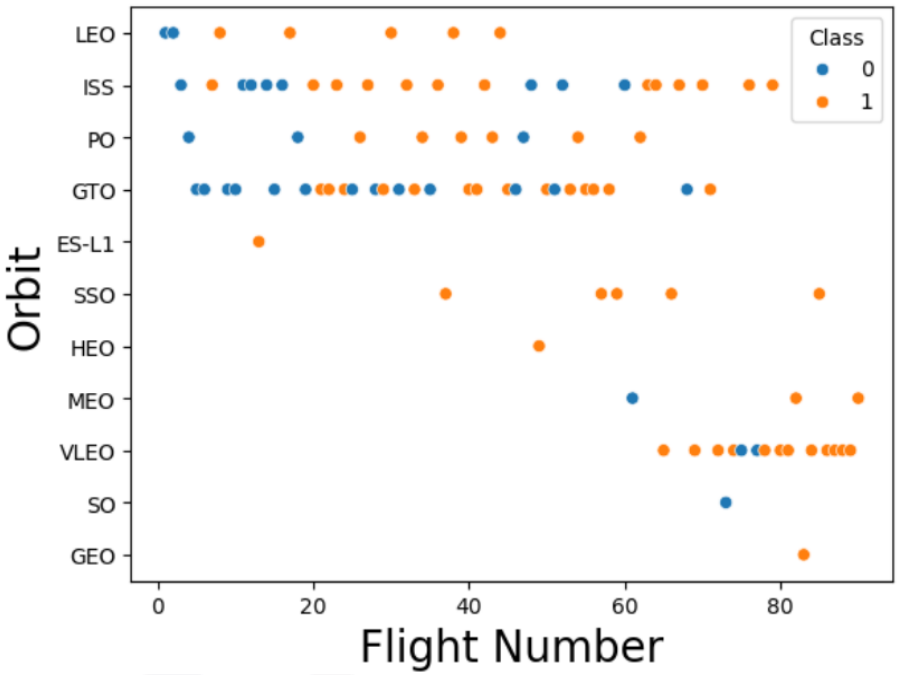
# RESULTS: Orbit Type Success Rates & Flight History

Takeaway: ES-L1, GEO, HEO & SSO had a 100% success rate.

SO had the lowest success rate of 0%




Bar Chart Showing Success Rate of Different Orbit Types




Scatter Plot Showing Success Rate against Flight Number

Takeaway: Flight Number Incraess Success Rate Regardless of Orbit Type (in most cases)

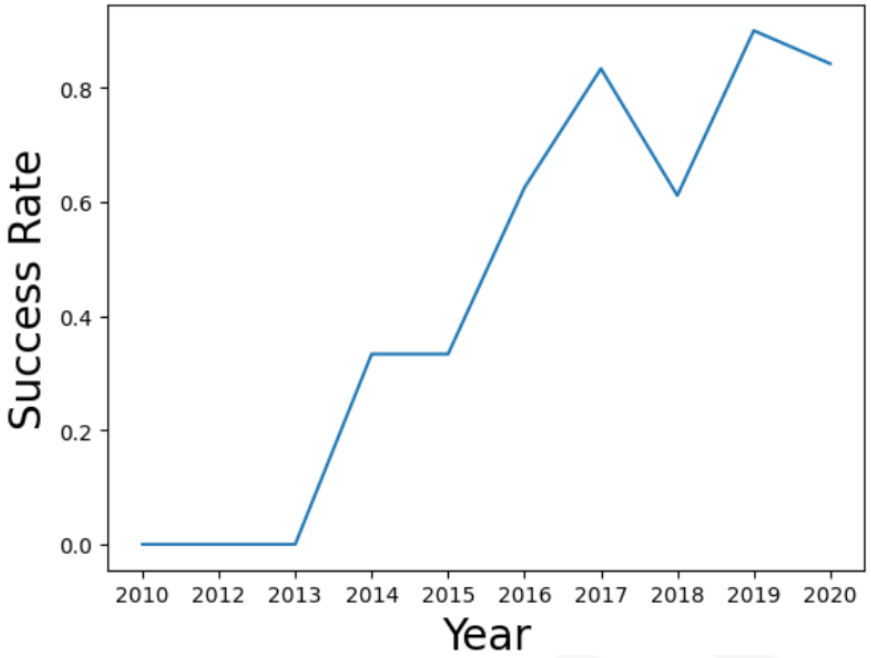


IBM

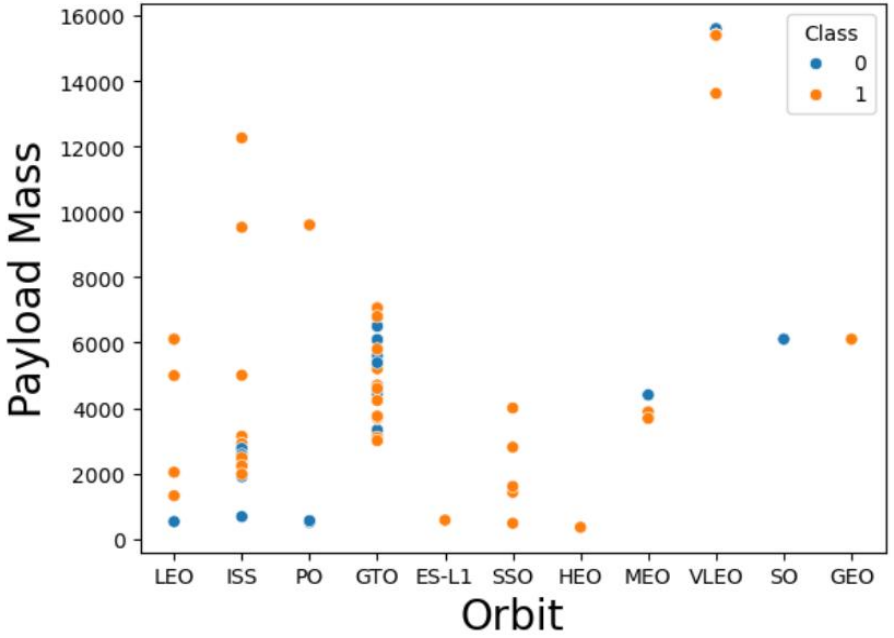
# RESULTS: Overall Success Trends & Payload Mass by Orbit

Takeaway: Success Rate Significantly Incraesed from 2010-2020.

Sharpest Increases were in 2013, 2015 & 2016




Line Chart showing Success Rate of Launches from 2010-2020

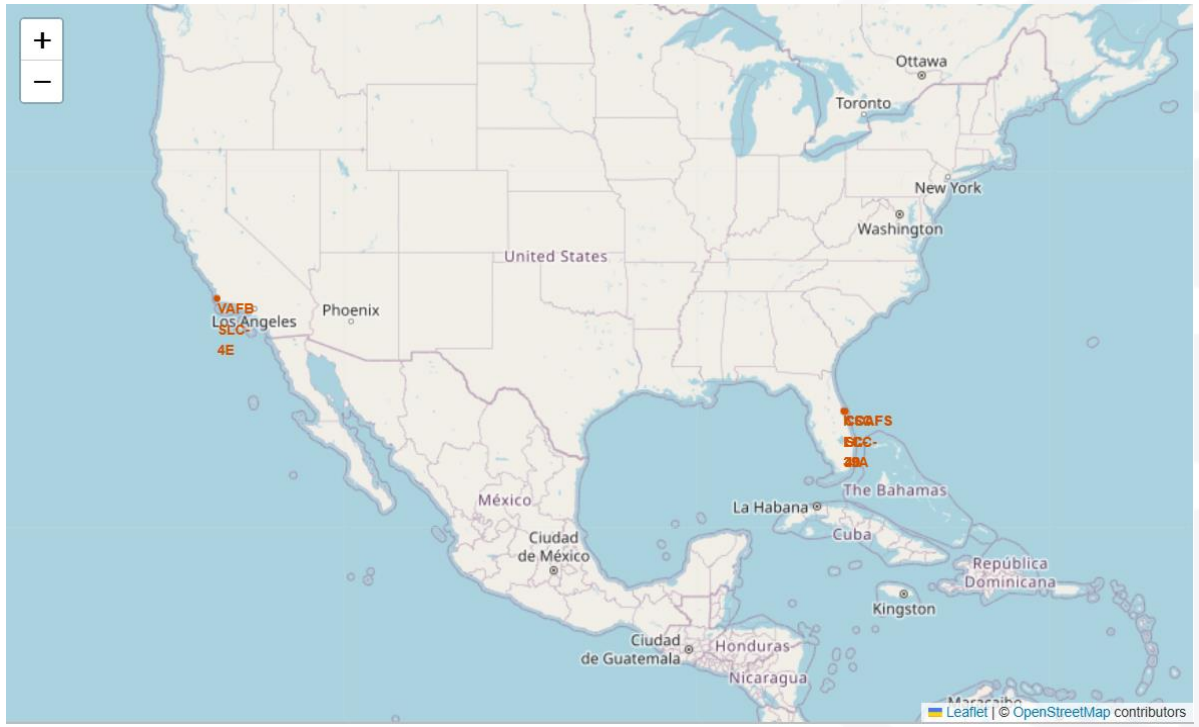



Scatter Plot showing Successes & Failures of Orbit Types versus Payload Mass (kg)

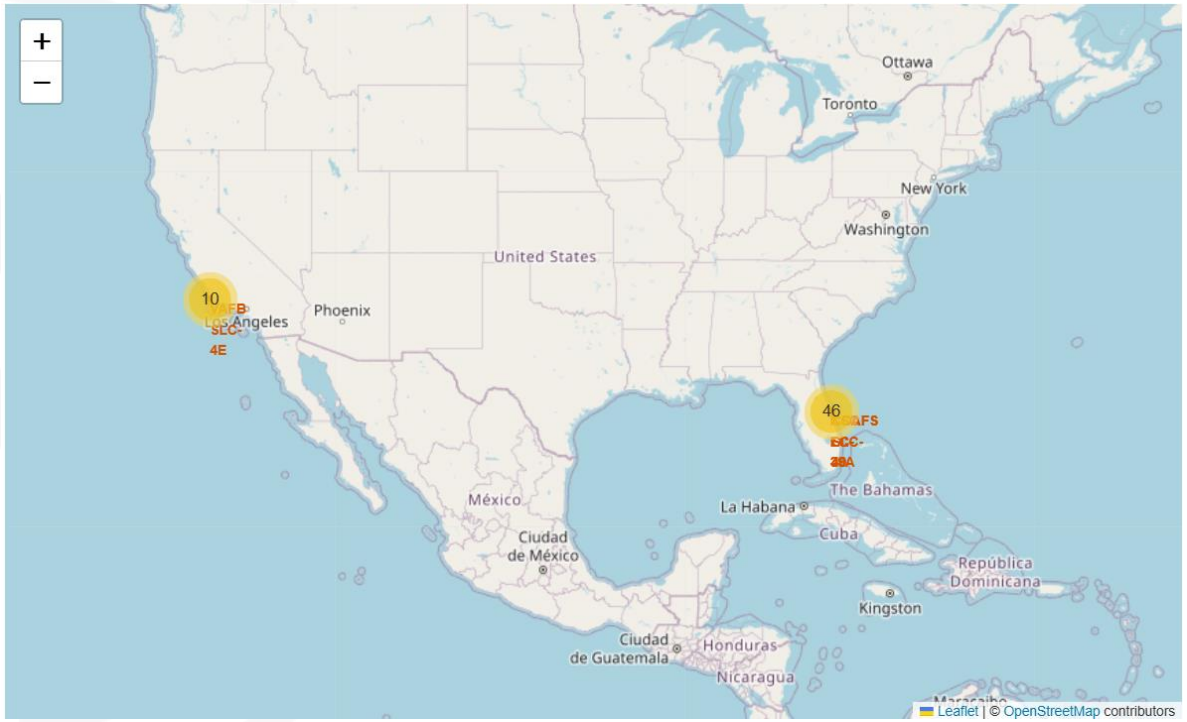
Takeaway: Success of Launches for Different Orbit Types varies with Payload Mass (kg)

# RESULTS: Maps with SpaceX Launch Sites in North America

Takeaway: Optimal Launch Sites are Close to the Coast (for safety), and/or the Equator (Orbital Mechanics)

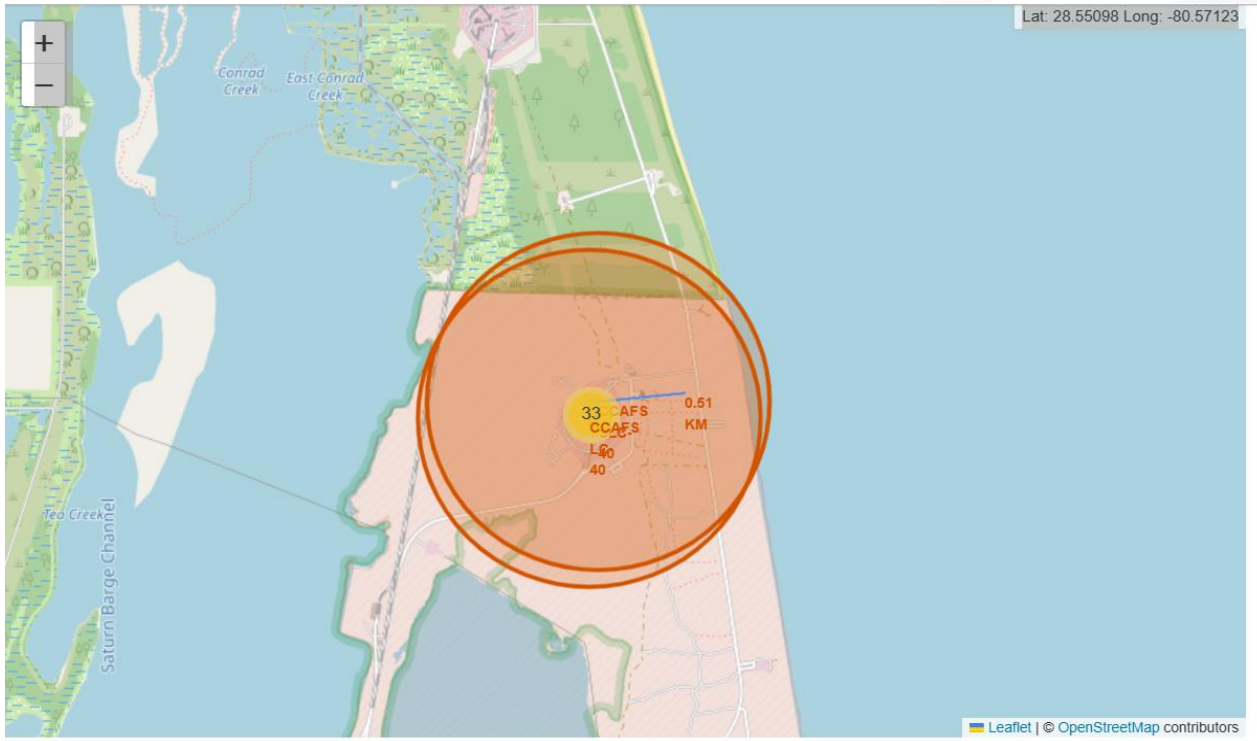



Map with Markers of SpaceX Launch Sites

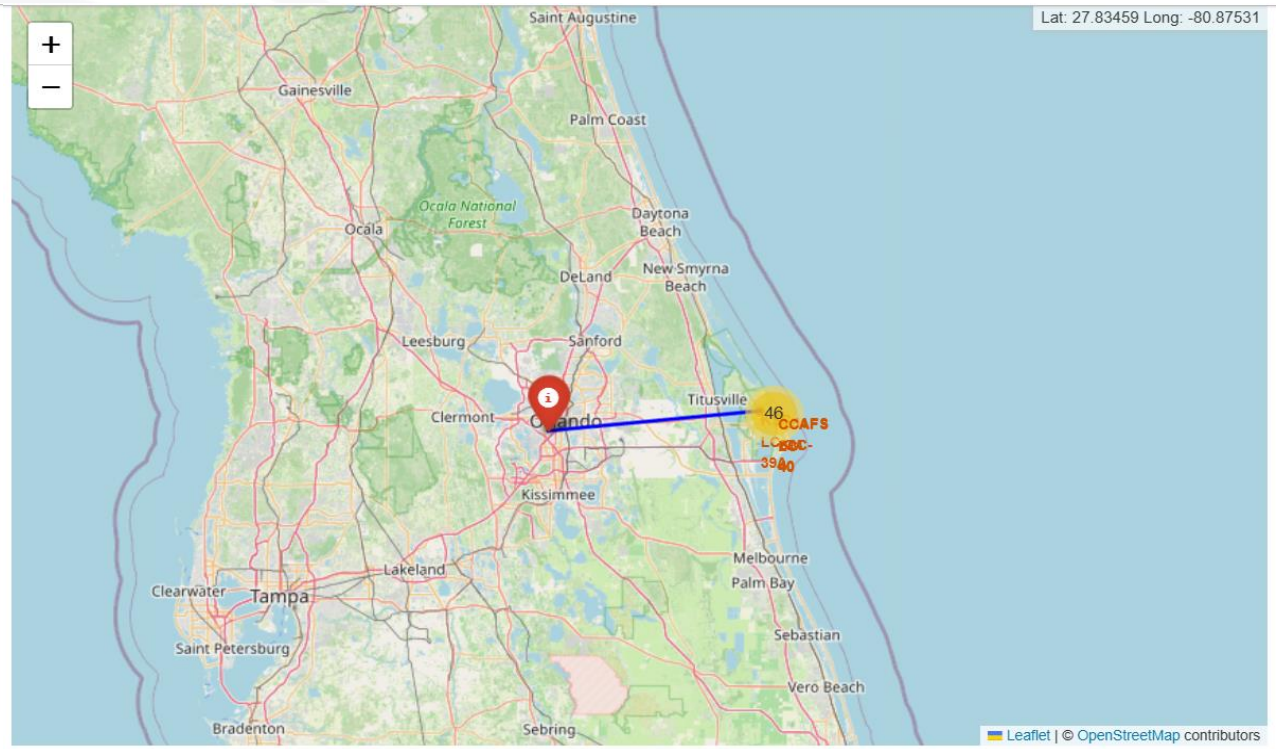



Map with Markers of SpaceX Launch Sites and Number of Launches at each Launch Site

# DATABASE TRENDS - FINDINGS & IMPLICATIONS



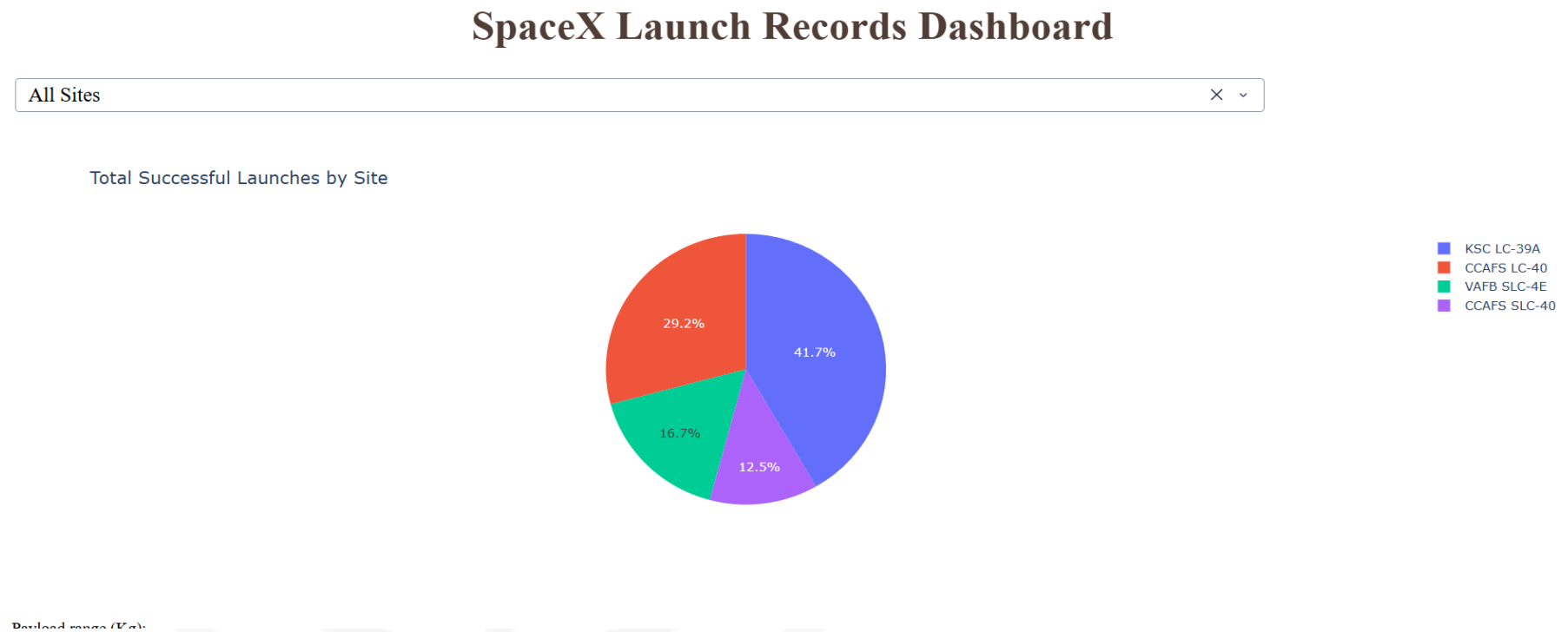

Map of SpaceX Launch Sites with Line to the Coastline




Map of SpaceX Launch Site with Line to the Closest City (Orlando)

# DASHBOARD: Pie Chart of Successful Launches at Each Launch Site



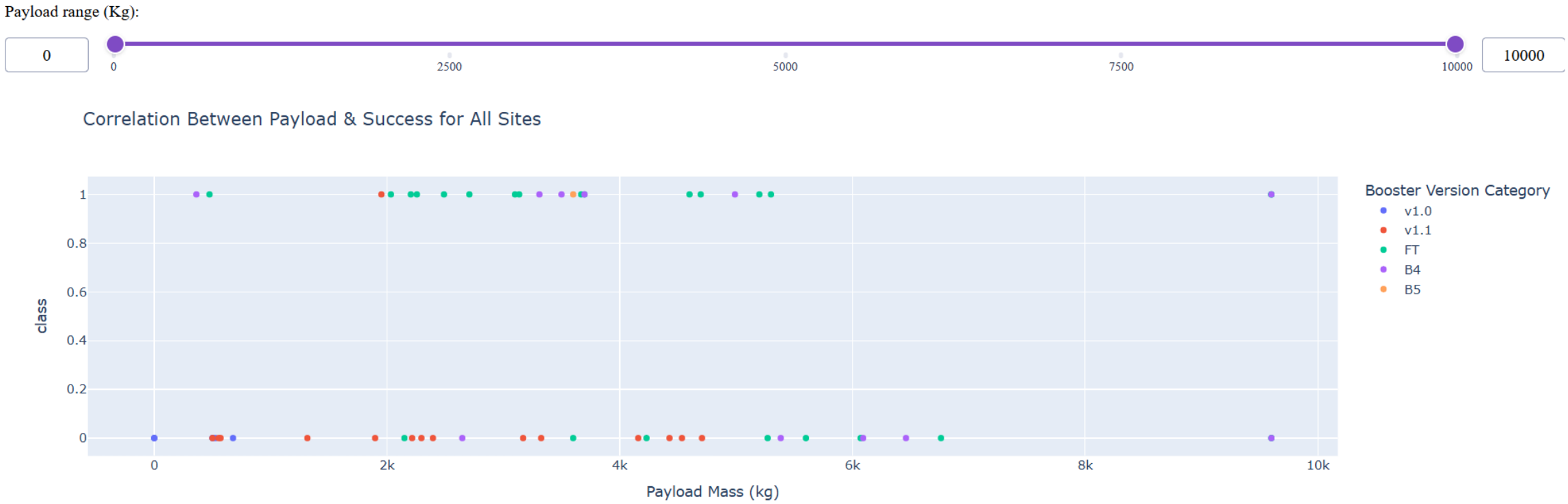



Pie Chart from Dashboard showing Pie Chart of Successful Launches by Launch Site

Includes a Dropdown Menu to Highlight Specific Launch Sites and Interactive Pie Chart that highlights specific statistics

# DASHBOARD: Scatter Plot of Payload & Booster Version

Takeaway: Allows Stakeholders to filter data by Payload Mass (kg) and Orbit Type in real-time to explore different launch scenarios.




Scatterplot from Dashboard Showing Payload Mass (kg) vs Booster Version

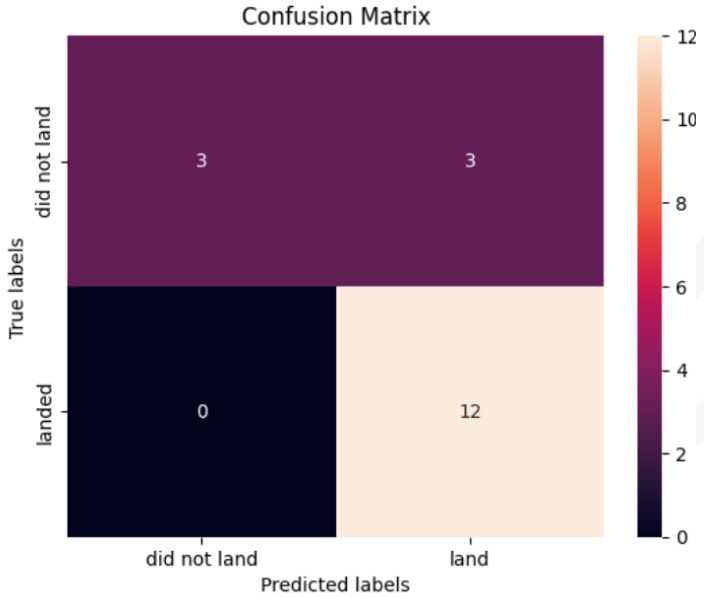
Includes interactive range slider and interactive scatter plot points

# PREDICTIVE ANALYSIS: Machine Learning Model Results

| Model | Training Accuracy | Test Accuracy |
|---|---|---|
| Logistic Regression | 84.64% | 83.33% |
| Support Vector Machine | 84.82% | 83.33% |
| Decision Tree | 86.07% | 83.33% |
| K-Nearest Neighbours | 84.82% | 83.33% |

Table showing Machine Learning Model Training & Testing Accuracy

All Machine Learning Models had the Same Confusion Matrices (12 Successes, 3 Failures, 3 False Positives, 0 False Negatives




Takeaway: All four models were competitive in Test Accuracy, but Decision Tree comes out on top due to its superior cross-validation Training Score (86.07%)

# DISCUSSION



- Implications: Reliable landing predictions to improve cost estimation & bid pricing.
- Launch Site Selection: Proximity to Coast/Equator & historical success should guide locations of future sites.
- Key Predictors:
  - Launch Number
  - Launch Site
  - Payload Mass
- Limitations:
  - Small sample size
  - Missing Payload Mass
  - Development of Technology over Time
- Business Value: Better predictors reduce financial risk for SpaceX and clients.

# CONCLUSION



- Conclusion: We predicted the Falcon 9 first-stage landing success via API & Web Scraping and ML.
- Outcome: Decision Tree Model achieved 86.07% Training Accuracy; Launch Site & Number are key factors.
- Future Work:
  - Expanded dataset
  - Add Weather
  - Deploy Real World Dashboard
- Final Message: Data-driven insights can optimise launch decisions and reduce costs.

# APPENDIX



- SpaceX APIs Data Sources:
  - getBoosterVersion: Rockets; URL: https//:api.spacexdata.com/v4/launches/rockets/
  - getLaunchSite: Launchpads; URL: https//:api.spacexdata.com/v4/launches/launches/
  - getPayloadData: Payloads; URL: url: https//:api.spacexdata.com/v4/launches/payloads/
  - getCoreData: Core; URL: https//:api.spacexdata.com/v4/launches/cores/
- Web Scraping Sources:
  - https://en.wikipedia.org/w/index.php?title=List_of_Falcon_9_and_Falcon_Heavy_launches&oldid=1027686922
- Tools: Python, Pandas, Folium, Plotly Dash, Scikit-Learn